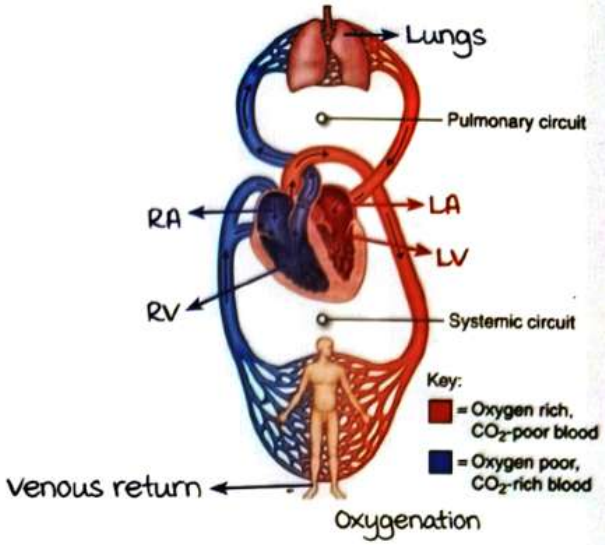
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CARDIAC OUTPUT

Cardiac output (CO) = Heart rate (HR) x stroke volume (SV).

- Sympathetic stimulation : ↑HR → ↑CO.
- Parasympathetic stimulation : ↓HR → ↓CO.

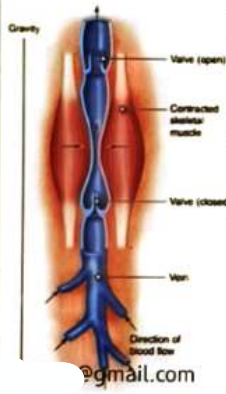
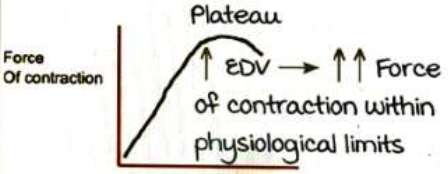
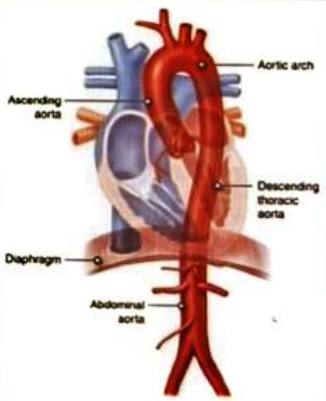


Determinants of Stroke Volume

00:03:27

FACTORS AFFECTING STROKE VOLUME

- Mnemonic (CAP).

	Preload	Contractility (Inotropy)	Afterload
Determined by	End diastolic volume (EDV) Venous return (VR)	Ejection fraction (EF)	Total peripheral resistance (TPR) Aortic pressure
Best marker	EDV	EF	TPR
Features	 Factors ↑ VR 1. Calf muscle contraction. 2. One way valve. 3. Deep fascia (Prevents collapse of veins).	 Frank starling law : (Length-tension relationship)	
Relation with CO	∝ CO	∝ CO	∝ 1/CO
Applied aspect	–	Digoxin • Positive inotropic agent (↑contractility → ↑EF). • Used in Rx of heart failure.	Heart & kidney form parallel circuit ↓ Removal of kidney ↓ ↑ TPR, (↑ Afterload) ↓ CO

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Factors affecting force of contraction :

- Bowditch effect/force-frequency relationship/treppe.
  ↑ HR → ↑ Force of contraction.
- Anrep effect : Abrupt ↑ in afterload → ↑ Force of Contraction.

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Measurement & Distribution 00:18:44

MEASUREMENT

1. Fick's principle :
   - O2 consumption = Arteriovenous oxygen difference x blood flow.
     - Cardiac output = $\frac{\text{O2 consumption}}{\text{AVO2 difference}}$ = 5 L/min.
     - Heart has highest AVO2 difference = 114 mL/L.
2. Echocardiography (m/c) : Clinically used.
3. Thermodilution method
4. Dye dilution method

} Theoretical.

DISTRIBUTION

Distribution & Oxygen Consumption :

Organ	Distribution	O2 Consumption
Liver	1500 mL/min (Highest)	51 mL/min (Highest)
Brain	750 mL/min	46 mL/min
Kidney	1250 mL/min 420 ml/100g/min (Highest)	-
Resting skeletal muscle	3mL/100g/min	-
Heart	250 mL/min (5% of CO)	10 mL/100g/min (Highest)

Note :

Cardiac Index Ratio : Cardiac output per m^2 body surface area.

Normal Cardiac Index ratio = 3.2L/min/m^2 BSA.

Coronary Circulation 00:28:53

Heart receives majority of blood during diastole.

Subendocardial part : more prone to ischemia.

Myocardial oxygen demand ∝ HR, duration of systole, systolic BP.

REGULATION OF CORONARY BLOOD FLOW :

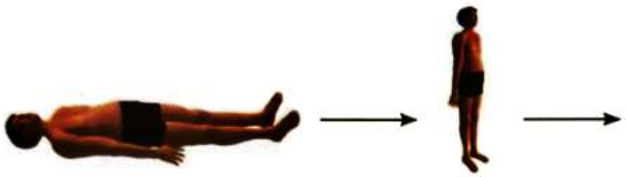
1. Nervous control : Minimal role.
2. Local factors : Accumulation of adenosine, NO, H+, metabolic end product → ↑ coronary blood flow.

Feedback

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3. Postural change :



Supine → On immediate standing

Transient fall in BP (BP = CO x TPR)

- Parameters affected : VR ↓, CO ↓, TPR ↑, HR ↑, Central venous pressure ↓.
- mechanism maintaining BP :
  - Baroreceptor reflex
  - ↑ TPR (To counter ↓ CO).

Shock

00:38:38

TYPES :

	Seen in	Cardiac Output
Hypovolemic (m/c)	RTA	↓
Neurogenic	General / Spinal anesthesia	↓ (D/t Sympathetic inhibition → Extensive vasodilation → ↓ TPR)
Cardiogenic	Cardiac disorders Eg: Cardiac tamponade	↓
Anaphylactic	Bee sting	↓ (D/t histamine release → ↑ capillary permeability → Extensive vasodilation)
Septic	Bacterial infection	↑ (D/t Septicemia → ↑ body temperature → Activation of SA node)

Hypovolemic Shock :

Stages :

	1. Non Progressive stage	2. Progressive stage	3. Irreversible stage
Feature	Compensatory mechanism present	Without Rx → Death	Death inevitable
MOA	• Sympathetic activation ↓ Norepinephrine release – Heart : ↑ HR, ↑ contractility, ↑ CO, ↑ BP. – Kidney : Vasoconstriction of afferent arteriole ↓ Diverts blood to heart & brain • Antidiuretic hormone release : ↑ blood volume (D/t ↑ water reabsorption)	• ↓ coronary blood flow. ↻ Cardiac deterioration ↻ Positive feedback • Clot formation in small blood vessels : ↓ blood flow to heart.	• D/t loss of energy sources. • ATP – Adenosine regeneration; Slow process. • Creatinine phosphate.

Feedback